CASE STUDY 1 – Illustrative Solutions

AUDITING ENTITY-LEVEL CONTROLS

Note to Instructors: There are three exercises in the front portion of Case Study 1. The primary purpose of these exercises is to impress upon students the importance of understanding entity-level controls and how to audit them. The exercises, which are research-oriented, can be used at the discretion of the instructor to generate class discussion. Due to the purpose and nature of these exercises, we have not provided illustrative solutions for them.

The case study activities are presented below in ***bold italic*** print. Our illustrative solution for each activity is presented in regular print.

Scenario 1: Ethical Behavior is Good Business

An illustrative business objective and associated business risk that reflect the company's philosophy regarding ethical conduct are expressed as follows:

Business Objective: To demonstrate sound ethical conduct in everything we do.

Business Risk: Disregard for sound ethical principles, either intentional or unintentional, may cause managers and employees to cut corners, embellish performance results, misuse company resources, or otherwise act in a manner that harms the company and its stakeholders.

1. Management asserts that entity-level controls are designed adequately and operating effectively to reduce the above stated risk to an acceptably low level. Identify the key entity-level controls you expect to find in place if management's assertion is true. Keep in mind that entity-level controls may exist in any of the five COSO components of internal control (control environment, risk assessment, control activities, information and communication, and monitoring activities). [Note: Students are encouraged to review Exhibit CS1-4, but answers should be case-specific.]



Note to Instructors: Solutions to this activity will vary depending on how instructors and students interpret the Committee of Sponsoring Organizations of the Treadway Commission's (COSO's) definitions of the five components of internal control. The textbook authors believe it is much more important for students to consider all five components of internal control when they formulate their answers than it is for them to "correctly" categorize the controls they identify. It is the authors' contention that control activities and information and communication are not as common at the entity level as control environment, risk assessment, and monitoring activities.

Key entity-level controls internal auditing should expect to find in place include, for example:

- Control environment:

  - Senior management fosters a strong corporate ethical climate by what they say and what they do as reflected in the slogan, "ethical behavior is good business," and the business objective, "to demonstrate sound ethical conduct in everything we do."
  - Senior management leads by example when faced with tough business decisions involving ethical ramifications.
  - The company has a comprehensive code of conduct that is applicable to directors, management, and employees.
  - Performance evaluation standards include ethical conduct criteria.
  - Actions that positively reflect sound ethical values are rewarded and appropriate disciplinary actions are taken in response to ethical misconduct.
- Risk assessment:
  - The board and senior management have established and clearly communicated a "zero tolerance" threshold for ethical misconduct.
  - Senior management regularly reassesses the inherent risk of ethical misconduct, taking into consideration external and internal changes that could affect the behavior of personnel throughout the company.
- Control activities:
  - Escalation procedures for reporting deficiencies in controls over the company's ethics program and suspected violations of the code of conduct to appropriate levels of management have been defined in writing and communicated throughout the company.
  - The company requires that all personnel, including senior management and directors, certify in writing on an annual basis that they understand and are in compliance with the code of conduct.
- Information and communication:
  - Newsletters, webcasts, and meetings are used to continuously educate company personnel about the company's ethics program and remind them of the importance of sound ethical behavior.
  - Management establishes an anonymous hotline and reports significant issues to senior management and the audit committee on a timely basis.
  - The code of conduct is posted prominently on the company's website.
- Monitoring activities:
  - The board oversees senior management's ethical conduct.
  - Management reviews the reports from the anonymous hotline and analyzes these reports for indications of need for improvement in policies and training.
  - The ongoing design adequacy and operating effectiveness of the entity-level controls over the company's ethics program are assessed periodically. (Note: The monitoring of entity-level controls may be conducted by senior management, internal auditing, or both.)

2. ***Many elements of the control environment are "soft" in nature. They may, for example, involve senior management behavior that intrinsically leaves little, if any, audit trail. An example of a soft control that you may have included in your answer***

to question 1 above is expressed as follows:

Senior management fosters a strong corporate ethical climate by what they say and what they do. They lead by example when faced with tough business decisions involving ethical ramifications.

a. Identify the audit procedures you would use to determine whether this control exists within SHR Corporation (SHR). Be specific.

The internal audit procedures that could be used to determine whether this control exists within the company include, for example:

- Interviewing executives, asking them specifically how they incorporate ethical considerations into their decision-making, address ethical issues that come to their attention, and exemplify sound ethical behavior.
- Surveying representative samples of managers, employees, and third-party stakeholders regarding their perceptions of senior management's ethical conduct.
- Observing executives as they interact with company personnel during meetings, paying particular attention to how they address questions and issues involving potential ethical ramifications.
- Reading company documents such as the annual report and newsletters, looking specifically for statements that reflect senior management's philosophy regarding business ethics.

b. Assume that this control does in fact exist within SHR.

1) Identify the audit procedures you would use to determine whether it is operating effectively. Be specific. Keep in mind that you need to build a sufficient body of appropriate evidence to support a valid conclusion.

Note to Instructors: Conducting procedures such as those listed above to determine the existence of the control also will provide evidence regarding its operating effectiveness. Building a "body of audit evidence" is important when testing "soft" controls because individual procedures such as surveys, interviews, or observations rarely provide persuasive evidence by themselves regarding design adequacy and operating effectiveness. In addition, testing relevant "hard" controls provides corroborative evidence that the soft controls are operating effectively.

In addition to those listed in a. above, the audit procedures that could be used to determine whether the control is operating effectively include, for example:

- Calling the anonymous hotline, reporting a made-up violation of the code of conduct, observing how the call is handled, and following up to determine whether appropriate actions were taken to address the reported issue.
- Determining the percentage of annual compliance certifications that are completed by company personnel and following up to determine

whether appropriate remedial actions were taken for personnel who did not submit compliance certifications on a timely basis.

2) ***Describe the evidence you might find that would indicate operating effectiveness.***

Audit evidence indicating operating effectiveness might include, for example:

- Cogent responses from executives regarding the steps they take to establish and sustain a sound ethical culture.
- Documentary evidence in corporate publications of senior management's philosophy regarding ethical conduct.
- Forthright and transparent communications from executives regarding business decisions involving ethical ramifications.
- Positive survey responses from managers, employees, and third-party stakeholders that indicate respect for senior management.
- Positive results from testing the anonymous hotline.
- A high percentage of annual compliance certifications submitted by company personnel on a timely basis.

3. ***The Public Company Accounting Oversight Board's (PCAOB's) Auditing Standard No. 5 indicates that entity-level controls include both (1) controls to monitor other controls and (2) controls to monitor results of operations. Provide an example of each type of monitoring control that would be useful in mitigating the business risk expressed above.***

Internal auditing's monitoring of the anonymous hotline is an example of an entity-level control designed to monitor another control. Use of the hotline by company personnel to report legitimate instances of suspected wrongdoing provides audit evidence that the hotline was operating effectively. Timely reporting of the results of monitoring activities to appropriate levels of management and/or the audit committee is evidence that the monitoring control is operating effectively.

The compensation committee's periodic review of executives' incentive compensation is an example of an entity-level control designed to monitor performance. Another example is the audit committee's oversight of financial executives' decisions regarding related-party transactions, accounting estimates, and other financial reporting matters of a subjective nature.

4. **Note to Instructors:** The questions presented below and the corresponding answers are based on ***the*** scenario presented in activity 4, which is not repeated here.

a. ***Identify the strengths and weaknesses in SHR's entity-level controls over the company's ethics program. Be sure to consider both governance and management-oversight controls. Propose specific recommendations to rectify the***

weaknesses noted. Based on the information provided, formulate an overall conclusion about the effectiveness of SHR's entity-level controls over its ethics program.

The strengths in the company's entity-level controls over its ethics program include:

- The company has a comprehensive, written code of conduct (the Code) that is applicable to directors, management, and employees.
- The Code is posted on the company's website and intranet.
- All new employees receive a copy of the Code when they are hired and participate in orientation training that includes coverage of the Code.
- Company policy requires that all company personnel certify in writing on an annual basis that they understand and are in compliance with the Code.
- The company has a whistleblower program that includes an anonymous hotline for employees and third parties to report suspected violations of the Code.
- The vice president of internal auditing and her direct reports monitor the whistleblower hotline and report suspected violations of the Code to appropriate levels of management and/or the audit committee.
- Internal auditing recently surveyed senior management and the audit committee regarding the company's ethics program and conducted follow-up interviews with the audit committee chair and selected members of the senior management team.
- Internal auditing separately surveyed a sample of managers and employees regarding the company's ethics program.

The weaknesses in the company's entity-level controls over its ethics program include:

- No substantive revisions to the Code have been made since its inception three years ago. A weakness in entity-level controls exists if follow-up audit procedures indicate that the Code has not been subjected to rigorous periodic reviews by representatives of the senior management team. Recommendation: Senior management should review the Code annually and update it as necessary to ensure that it remains current, taking into consideration external and internal changes affecting the company.
- The audit committee's charter includes no provision for overseeing senior management's ethical conduct or monitoring its adherence to prescribed internal control policies and procedures. Recommendation: The audit committee should establish procedures for monitoring executives' ethical conduct.
- Senior management's assessment of the risk of ethical misconduct is not sufficiently robust. The assessed level of inherent ethical misconduct risk has remained consistently low over the past three years. This is inconsistent with the challenges the company is now facing—the economic downturn, increasing competition, and decreasing profitability—and indicates inadequate consideration of such factors during the entity-level risk

assessment process. Such challenges increase the inherent likelihood of managers and employees cutting corners and/or embellishing performance results. Recommendation: The board and senior management should acknowledge the fact that the new challenges facing the company have increased the inherent risk of ethical misconduct and determine whether existing internal controls are adequate to mitigate this higher level of risk.

- Twenty-five percent of the managers and employees who received internal auditing's survey regarding the company's ethics program did not respond. (Note: This fact may or may not represent a deficiency. If the response rate reflects apathy, or a feeling among managers and employees that no one really cares about their feelings, then it may be a deficiency. If instead the response rate reflects the fact that the survey is anonymous and personnel without strong feelings regarding the company's ethics program just didn't respond, then it may not be a deficiency.) Recommendation: Senior management should communicate its expectation of 100 percent participation in the survey and its expectation that supervisors strongly encourage their subordinates to participate.
- In 2012, 20 percent of managers and 10 percent of employees failed to comply with the company's policy that managers and employees certify in writing on an annual basis that they understand and are in compliance with the Code. Recommendation: Senior management should communicate its expectation of 100 percent compliance and supervisors should be held accountable for ensuring that their subordinates comply.
- Written performance evaluation standards include no ethical conduct criteria. No provision exists for rewarding personnel that demonstrate sound ethical behavior or for disciplining those that demonstrate ethical misconduct. Recommendation: Performance evaluation standards should include ethical conduct criteria and performance should be judged against these standards.
- Calls on the anonymous hotline have been infrequent, with very few suspected Code violations of a significant nature being reported to senior management or the audit committee. This may indicate that violations of the Code are in fact infrequent. On the other hand, it may mean that managers and employees are unenthusiastic about using the hotline because they see no evidence that appropriate actions will be taken to address unethical conduct. The hotline is of minimal value if appropriate actions are not taken in response to misconduct. A widespread perception among managers and employees that senior management "talks-the-talk" but does not "walk-the-walk" regarding ethical misconduct may undercut the entire ethics program. Recommendation: Senior management should continuously reinforce the importance of using the anonymous hotline and communicate actions that have been taken in response to instances of known ethical misconduct.
- Ethics training is not required and the company offers no ethics training in-house other than that provided to new employees. Recommendation: All company personnel should be required to complete a prescribed amount of relevant ethics training each year.
- Senior management's communication of the company's philosophy regarding ethical conduct and its ethics policies may be inadequate.

Managers and employees indicated in their survey responses that senior management does very little to inform them about the company's ethical policies. This perception is inconsistent with senior management's stated belief that it clearly conveys the importance of ethical conduct and compliance with the Code via e-mails, webcasts, and town hall meetings. Recommendation: Senior management should investigate this matter further to determine the reasons for the gap in perceptions and how to best close the gap.

Overall conclusion:

Internal auditing's assessment of the company's entity-level controls over its ethics program must take into consideration the fact that the inherent risk of ethical misconduct has increased due to the challenges the company is now facing: the economic downturn, increasing competition, and decreasing profitability. Taken together, the weaknesses identified above represent a significant deficiency in the company's entity-level controls that internal auditing needs to report to senior management and the audit committee.

b. Discuss how the entity-level controls relating to the company's ethics program might affect senior management's decisions given the nature of the risks the company now faces. Provide some examples.

The risks now facing the company put more pressure on senior management and there are no formal policies and procedures in place to monitor senior management's ethical conduct. Lack of proper oversight opens the door for senior management to make decisions that are not in the best interest of the company and its stakeholders. An executive might, for example, look for opportunities to manipulate performance measurements to make the company appear to be doing better than it really is.

c. Discuss how the entity-level controls relating to the company's ethics program might affect managers' and employees' behavior at the business process level. Provide some examples.

Evidence suggests that senior management may be preoccupied with the lackluster performance results the company is achieving and, therefore, not paying sufficient attention to the company's ethical culture. Senior management's distraction widens the window of opportunity for business process managers and employees to act unethically if they are so inclined. Business process managers and employees also may feel pressure to embellish performance results, especially if their compensation and continued employment depends on achieving designated performance targets.

d. Discuss how SHR's entity-level controls over the company's ethics program may impact your subsequent audit of controls over the company's purchases and accounts payable process. More specifically, discuss how your conclusion would affect your:

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1) ***Professional skepticism.***

Professional skepticism means keeping an open mind, critically assessing audit evidence, and gathering persuasive evidence. Under conditions of increased uncertainty due to extenuating circumstances such as those caused by the pressures the company is now facing, internal auditors must conduct the audit with a heightened degree of professional skepticism and take special care to make sure they gather sufficient appropriate evidence to support well-founded conclusions.

2) ***Assessment of process-level inherent risks.***

Inherent risk at the process level is the gross risk that exists assuming there are no process-level controls in place. Deficiencies in entity-level controls increase inherent risk at the process level. Internal auditing should assess the inherent process-level risks as being higher than they would be if a significant deficiency did not exist in the entity-level controls over the company's ethics program.

3) ***Approach to evaluating the design adequacy of process-level controls.***

Internal auditing's evaluation of process-level controls is influenced by the assessed level of inherent risk. Controls that are adequate under conditions of low inherent risk may be inadequate for reducing higher inherent risks to acceptably low levels. The increased level of inherent risk at the process level should prompt internal auditing to more rigorously test the design adequacy of controls.

4) ***Inclination to perform direct tests of transactions in search of indirect evidence to support your conclusion about the design adequacy and operating effectiveness of process-level controls.***

A conclusion that a significant deficiency exists in the company's entity-level controls would prompt internal auditing to more rigorously test controls at the process level and transaction level. Since direct tests of transactions provide indirect evidence regarding the design adequacy and operating effectiveness of controls, internal auditing will be more inclined to perform such tests. Finding actual errors in transactions would add to internal auditing's "body of evidence" and further support the conclusion that internal controls are designed inadequately and/or operating ineffectively to prevent or detect such errors. Finding no errors in transactions does not necessarily mean, however, that controls are designed adequately and operating effectively.

5) ***Assume that internal auditing concludes that the effectiveness of SHR's entity-level controls over its ethics program is significantly deficient, but not to the level of a material weakness. Prepare a scenario in which the combination of the deficiencies in the entity-level controls over the company's ethics programs and deficiencies in related entity-level controls***

could rise to the level of a material weakness. Hint: Consider entity-level controls pertaining to the company's hiring and compensation practices.

A combination of entity-level control deficiencies that could rise to the level of a material weakness is the significant deficiency in controls over the company's ethics program together with the following:

- The company's hiring practices include no provision for conducting background checks on prospective executives; and
- Executives' compensation is based heavily on narrow performance targets that are easy to manipulate.

Scenario 2: Using IT to Gain a Competitive Edge

1. ***SHR's senior management team understands the importance of aligning the company's IT strategies with its business strategies. Identify two types of IT strategic decisions senior management already has made or is likely to make in the foreseeable future. Clearly explain the linkage between these IT strategic decisions and SHR's business strategies.***

1.

SHR has elected to leverage the power of the Internet by using its website to sell its products. This use of information technology greatly enhances the company's efforts to increase its market share by expanding its direct sales globally.

SHR implemented electronic data interchange (EDI) with its primary vendors several years ago. Using EDI to streamline its purchasing process and maintain an uninterrupted flow of incoming inventory to its distribution centers and retail stores has resulted in operating efficiencies, which has had a positive effect on the company's competitive pricing strategy.

2. ***Sound decision making requires high-quality information.***

a. ***What information does senior management need to make informed IT strategic decisions?***

The information senior management needs to make informed IT strategic decisions includes, for example:

- The IT strategic options available and the alignment of these options with the company's business objectives and strategy.
- The expected benefits and costs of viable strategic options.
- The risks associated with the viable options and the company's risk response options.
- The technical, economic, legal, operational, and schedule feasibility of the viable strategic options.

b. ***Identify the entity-level controls you would expect to be in place to ensure that senior management has high-quality information upon which to base its IT strategic decisions.***

Entity-level controls in place to ensure that senior management has high-quality information upon which to base its IT strategic decisions might include, for example:

- Direct involvement of the board, or a board committee, in the IT strategic decision-making process and final approval of strategic decisions by the board.
- A high-level IT steering committee that includes executives and other individuals that collectively possess the necessary expertise to make sound strategic decisions.
- Asking internal auditing to provide assurance regarding the quality of the information upon which strategic decisions will be based and to provide advice regarding risks and risk response options.

3. ***Explain what role, if any, internal auditing should have in the IT strategic decision-making process.***

Internal auditing should not make IT strategic decisions—such decisions are senior management's responsibility. Internal auditing can, however, assist in the decision-making process by providing management with assurance that the information upon which a strategic decision will be based is, in fact, sufficient, relevant, reliable, accurate, and timely. Internal auditing also can provide senior management with information of an advisory nature such as the risks associated with a particular strategy and the risk response options that should be considered.

4. ***One of SHR's business strategies is to selectively acquire companies that complement its core competencies.***

   a. ***Explain the effects a business acquisition could have on the inherent risk of failure to effectively and efficiently integrate acquired IT resources into its business processes.***

   A business acquisition could increase the inherent risk of failure to effectively and efficiently integrate acquired IT resources into SHR's business processes for reasons such as:

   - The acquired company may have information systems that are incompatible with SHR's information systems.
   - The information systems of the combined company may be distributed over a wide geographical area, including systems in different countries.
   - The personnel in the acquired company may be resistant to change and/or lack the expertise required to assist in the integration process.

   b. ***Describe the entity-level controls SHR should have in place to mitigate these effects.***

   The entity-level controls SHR should have in place to mitigate these effects include, for example:

- Procedures built into the acquisition due diligence process that will provide SHR with the information it needs regarding the information systems, technology expertise, and cultures of companies being considered for acquisition.
- A strategy for imbedding the SHR technology and change management expertise needed to align the information systems of acquired companies with SHR's information systems.

5. ***SHR uses EDI to purchase inventory from its primary vendors.***

   a. ***Identify the inherent business risks associated with the use of EDI. Include in your answer both entity-level risks and process-level risks.***

   Entity-level inherent risks associated with SHR's use of EDI include, for example:
   - Business interruptions caused by natural disasters such as fires, floods, or earthquakes, or by hardware or software system failures.
   - Failure to take strategic advantage of advances in EDI systems. For example, continuing to use a third-party value added network (VAN) when migrating to a newer Internet/intranet system would be more cost effective.
   - Partnering with vendors that subsequently prove to be unreliable.
   - Unauthorized access by hackers to proprietary information resulting in misuse, destruction, or theft of the information.
   - Virus and worm attacks.

   b. ***Identify the governance-level controls associated with the company's use of EDI you would expect to find in place.***

   Process-level inherent risks associated with SHR's use of EDI include, for example:
   - Interception or corruption of procure-to-pay transaction information during transmission to or from vendors.
   - Repudiation of purchase or payment transactions by vendors.
   - Unauthorized access by vendors to proprietary information unrelated to the procure-to-pay process.
   - Transmission of invalid, inaccurate, or incomplete data by vendors.

   c. ***SHR's governance-level controls associated with its use of EDI might include, for example:***
   - Oversight and direction of the IT strategic decision-making process by the board and senior management.
   - Written policies regarding the conditions that must exist and the standards that must be met to establish EDI partnerships with vendors.
   - Monitoring compliance with the EDI partnership policies.

   d. ***Management asserts that controls are designed adequately and operating effectively to mitigate the risk of unauthorized individuals accessing the company's EDI system and misusing or misappropriating proprietary information or other assets.***

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1) ***Identify the management-oversight level IT controls you would expect to find in place if management's assertion is true.***

The IT management-oversight level IT controls internal auditing should expect to find in place include, for example:

- Installing a firewall that filters out and captures suspicious incoming data transmissions and taking appropriate follow-up actions to remediate suspicious transmission attempts captured by the firewall filter.
- Using authentication procedures that preclude hackers and other unauthorized individuals from accessing the company's information systems.
- Installing a high-level authorization table that specifies the systems and information that valid users can access and the activities they can perform once they have gained access to the systems and information.
- Conducting vulnerability assessments and penetration tests to identify system weaknesses that hackers and others can exploit to gain unauthorized access.
- Implementing intrusion detection software that alerts IT personnel to unauthorized access, monitoring the incidents captured by the software, and performing appropriate remediation procedures.

2) ***Identify the process-level controls and transaction-level controls, both manual and automated, you would expect to find in place if management's assertion is true.***

The manual controls internal auditing should expect to find in place include, for example:

- Updating the list of authorized users and the authorization table as personnel changes occur.
- Monitoring the access and activity log and taking appropriate actions to remediate suspicious access attempts and activities.

The automated controls internal auditing should expect to find in place include, for example:

- Using log-in procedures, including user identification and passwords, that restrict access to the purchasing application system to authorized users.
- Installing an authorization table that specifies the activities that valid users of the application system can perform once they have accessed the system.
- Logging all access attempts to the purchasing application system and the activities performed by system users.

3) ***Assume that internal auditing examined the IT management-oversight level***

controls identified ***in 1) above and found them to be inadequately designed. Explain how, if at all, this finding will affect the nature, timing, and extent of the audit procedures performed to determine the design adequacy and to test the operating effectiveness of the controls identified in 2) above. Provide specific examples to support your explanation.***

IT management-oversight level controls are implemented at a higher level and more pervasively to prevent or detect timely unauthorized access to the company's information systems as a whole. Design deficiencies in this layer of controls increase the risk of unauthorized access at the process level. Example: Vulnerabilities in the company's firewall that allow a hacker to penetrate the company's outside layer of information system protection puts the hacker one large step closer to gaining unauthorized access to specific application systems.

The process-level and transaction-level controls designed and implemented to reduce the risk of unauthorized access to the system will need to be stronger to compensate for the vulnerabilities in the company's firewall and to provide reasonable assurance that the residual risk is reduced to an acceptably low level. This in turn means that internal auditing will need to more rigorously test the design adequacy and operating effectiveness of the process-level and transaction-level controls to obtain sufficient evidence supporting a conclusion that the residual risk of unauthorized access to the purchasing application system is acceptably low.

6. ***All of SHR's purchasing processes, including those in which EDI is not implemented, are highly automated. SHR's perpetual inventory system also is fully automated.***

   a. ***Assume that internal auditing examined the IT general controls pertaining to SHR's purchasing processes and perpetual inventory system and found them to be designed adequately and operating effectively. Explain how, if at all, internal auditing's assessment of IT general controls will affect the nature, timing, and extent of the audit procedures performed to determine the design adequacy and to test the operating effectiveness of the IT application controls built into the automated purchasing processes and perpetual inventory system. Provide specific examples to support your explanation.***

   Business risks are mitigated by a combination of entity-level controls, which include IT general controls, and process and transaction-level controls, which include IT application controls. If *key* general controls pertaining to SHR's purchasing processes and perpetual inventory system are designed adequately and operating effectively, less reliance will need to be placed on related application controls to provide reasonable assurance that risks are reduced to an acceptably low level. Therefore, less rigorous internal audit testing of the IT application controls built into the automated purchasing processes and perpetual inventory system will be required than otherwise would be the case.

Example: Internal auditing tested *key* IT application controls last year and found them to be adequately designed and operating effectively. If internal auditing tests *key* IT general controls over application program changes (that is, IT change controls) on an ongoing basis and determines that they continue to be designed adequately and operating effectively, then internal auditing need not repeat the testing of the application controls.

This is especially true if internal auditing knows that the application controls have not changed since last being tested.

b. Assume that internal auditing found the IT application controls built into the automated purchasing processes and perpetual inventory system to be designed adequately and operating effectively. Explain how, if at all, this assessment will affect the nature, timing, and extent of the audit procedures performed to determine the design adequacy and to test the operating effectiveness of the manual process-level and transaction-level controls pertaining to purchases, accounts payable, and inventory. Provide specific examples to support your explanations.

Internal auditing should focus its testing on *key* process-level and transaction-level controls, which typically include both automated and manual controls. The testing of key manual controls will not be affected by internal auditing's conclusions regarding key automated controls because both types of controls are needed to reduce residual risks to acceptably low levels.

7. Management asserts that controls are designed adequately and operating effectively to mitigate the risk of delays in processing purchase transactions and inventory shortages caused by disruption or corruption of electronic transmissions between SHR and its EDI vendors.

a. Identify the management-oversight level IT controls you would expect to find in place if management's assertion is true.

The IT management-oversight level controls internal auditing should expect to find in place include, for example:

- Business continuity and disaster recovery planning that includes redundant web servers and data.
- Installing intrusion and virus detection software. (Note: System intruders or viruses can disrupt or corrupt data transmissions in addition to causing other problems.)

b. Identify the process-level controls and transaction-level controls, both manual and automated, you would expect to find in place if management's assertion is true.

The manual purchase transaction controls internal auditing should expect to find in place include, for example:

- Reviewing computer generated exception reports and taking timely

corrective action when sent and received control totals do not match.
- Verifying that acknowledgements sent by the receiving computer are received by the sending computer and taking appropriate actions when acknowledgements are not received when expected.

The automated controls internal auditing should expect to find in place include, for example:
- Computer matching of control totals when transmitted data is received and the generation of exception reports when control totals do not match.
- Transmitting a receipt acknowledgement to the sender.

c. Assume that internal auditing examined the IT management-oversight level controls identified in a. above and found them to be designed adequately and operating effectively. Explain how, if at all, this finding will affect the nature, timing, and extent of the audit procedures performed to determine the design adequacy and test the operating effectiveness of the controls identified in b. above.

Both the IT management-oversight level controls listed in a. above and the process-level and transaction-level controls listed in b. above are *key* to mitigating the risk of delays in processing purchase transactions and inventory shortages caused by disruption or corruption of electronic transmissions between SHR and its EDI vendors. They complement each other by addressing different aspects of the risk. Whereas the management-oversight level controls address problems that could occur at the front end or back end of data transmissions, the process-level and transaction-level controls address problems that could occur as data is traveling from one end to the other. In this situation, internal auditing's conclusion about the IT management-oversight level controls should have little, if any, effect on the testing of the process-level and transaction-level controls. (Note: The manual and automated controls identified in b. above are also complementary in nature. For example, having the computer generate exception reports when control totals do not match is of little value if no one reviews the reports or if someone reviews the reports but fails to take corrective action. Internal auditing needs to consider this fact when designing its tests—both the automated controls and the complementary manual controls should be tested to some extent.)

8. Assume that internal auditing, prior to beginning its testing of IT controls, concluded that deficiencies in entity-level controls over the company's ethics program and hiring and compensation practices, when considered in combination, constitute a material weakness in internal control. Explain what impact, if any, this conclusion should have on internal audit testing of IT controls at the:

a. Governance level.

The impact that internal auditing's conclusion should have on its testing of IT controls at the governance level will depend on whether the control deficiencies raise concerns about the ethical conduct and proficiency of the senior management team. Concerns about the ethical conduct and proficiency of senior management

would make it virtually impossible to audit IT controls at the governance level because the company's IT strategy is inseparable from its business strategy. The risk would be too great that senior management is making IT-related decisions that are not in the best long-term interests of the company and its stakeholders. Conversely, internal auditing's conclusion should have little or no impact on its testing of IT controls at the governance level if the ethical conduct and proficiency of senior management is not called into question.

b. Management-oversight level.

The impact that internal auditing's conclusion should have on its testing of IT controls at the management-oversight level will depend on whether the control deficiencies raise concerns about the ethical conduct and proficiency of the senior management decision-making team. Significant concerns about the ethical conduct and proficiency of senior management would make it relatively pointless to test management-oversight level controls

If internal auditing is not concerned about the ethical conduct and proficiency of senior management, they must be concerned about the next level of management—otherwise, they would not have concluded that the deficiencies in entity-level controls constitute a material weakness. The question then becomes whether internal auditing's concerns extend to IT management. Concerns about the ethical conduct and proficiency of IT management will have a significant impact on the manner in which internal auditing tests IT management-oversight level controls. They will need to greatly elevate their level of professional skepticism and much more rigorously test the design adequacy and operating effectiveness of the controls. Moreover, they will need to be constantly aware that there is a high risk of ethical misconduct and/or insufficient expertise within the IT management team.

c. Process and transaction levels.

The impact that internal auditing's conclusion should have on its testing of IT controls at the process and transaction levels will depend on the results of the testing conducted at the management-oversight level. The effectiveness of IT controls at the process and transaction levels is largely dependent on the effectiveness of higher-level IT controls. Ultimately, internal auditing will need to identify *key* controls, which are likely to reside at both the management-oversight level and at the process and transaction levels. Deficiencies in key controls at the management-oversight level may cause key controls at the process and transaction levels to be inadequately designed. Testing the operating effectiveness of inadequately designed controls typically makes no sense. On the other hand, key process and transaction controls may be adequately designed despite deficiencies in management-oversight level controls, in which case it makes sense to test their operating effectiveness. Ultimately, internal auditing needs to determine whether key controls at the management-oversight, process, and transaction levels collectively reduce residual risks to acceptably low levels.